ModelIt!

Student Activity Pack

Why Is Gold So Different from Iron?

How Atomic Structure Determines the Properties of Every Material

Grade 08L2 | NGSS MS-PS1-1, MS-PS1-3 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What makes one element different from another at the atomic level?

2 Why do some metals rust easily while others stay shiny for centuries?

3 What is an electron, and how might its position in an atom affect the atom's behavior?

4 Draw a simple atom and label the protons, neutrons, and electrons. How might adding more protons change the element?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Atomic Number | Electron Configuration | Reactivity | Conductivity | Material Properties

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gold vs. Iron

Atomic Number 79 vs. 26

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Alkali Metal

1 outer electron configuration

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Full Outer Shell

Complete electron shell configuration

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Melting Point — The temperature at which a solid becomes a liquid — related to the strength of bonds between atoms

Electronegativity — How strongly an atom attracts electrons in a chemical bond — affects bonding type and reactivity

Density — Mass per unit volume — heavier atoms and tighter packing increase density

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do two elements made of the same basic particles — protons, neutrons, and electrons — behave so differently? Use evidence from your simulation to support your answers.

1. **Why is gold so valued throughout human history — is it just appearance, or do its atomic properties play a role?**

2. **How does the periodic table organize elements by atomic structure, and what patterns emerge?**

3. **Why are noble gases so unreactive while alkali metals are so reactive?**

4. **How does understanding electron configuration help engineers choose materials for specific jobs?**

5. **If you could design a new element with custom properties, what electron configuration would you choose and why?**

8 STEM Challenge: Design a Material Selection Guide

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YOUR ENGINEERING MISSION

Design an evidence-based guide that recommends the best element or alloy for specific engineering applications, using your model data on atomic properties to justify each recommendation.

THE SCENARIO

An engineering company needs to select materials for three products: electrical wiring, a corrosion-resistant ship hull, and a lightweight aircraft frame. Your team must recommend materials based on conductivity, reactivity, and material properties.

Engineering Design Thinking

1. WHAT SPECIFIC ATOMIC PROPERTIES DOES EACH APPLICATION REQUIRE?
2. HOW DOES REACTIVITY AFFECT THE LONG-TERM DURABILITY OF EACH MATERIAL?
3. WHAT TRADE-OFFS EXIST BETWEEN CONDUCTIVITY, COST, AND CORROSION RESISTANCE?
4. HOW WILL YOU USE MODEL EVIDENCE TO JUSTIFY EACH MATERIAL RECOMMENDATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Scientists and Metallurgical Engineers study how atomic structure determines material properties and design new materials for technology, medicine, and construction. They earn $75,000-$130,000/year.